Unit 1 Database management systems

Work through this revision selection before checking the separate answer book. The full lesson and exercise bank is available on StudyLab.

SQL schema: Students(StudentID, StudentName, City, Age); Courses(CourseID, CourseName, Fee); Enrolments(EnrolmentID, StudentID, CourseID, Mark).

1.1 Compare database users

A college has a lecturer entering marks, a developer building the portal and a DBA restoring a backup. Explain each responsibility.

Your answer

1.2 Plan a database

A shop keeps conflicting customer phone numbers in three spreadsheets. Propose a database plan and two migration checks.

Your answer

1.3 Evaluate costs

Give two costs of replacing shared spreadsheets with a managed database, and explain why a backup alone is not a recovery plan.

Your answer

Unit 2 Data models and ER diagrams

Revision selection Original supplementary material by Lefa Malatji

2.1 Read cardinality

State the cardinalities when each ticket has one requester and each requester may create many tickets.

Your answer

2.2 Model optional participation

A ticket may be unassigned or assigned to one technician; each technician may have many tickets. Name the FK and nullability.

Your answer

2.3 Choose attribute types

Classify a person name, date of birth, age and multiple phone numbers. Explain a suitable relational design.

Your answer

Unit 3 Relational design and keys

Revision selection Original supplementary material by Lefa Malatji

3.1 Choose candidate keys

A student has StudentID, an institutionally unique StudentNumber, Name and City. Identify candidate, primary and secondary/search keys.

Your answer

3.2 Locate a foreign key

An order belongs to one customer. Explain why the customer key belongs in Orders rather than storing a list of order IDs in Customers.

Your answer

3.3 Revise an ERD

An employee previously belonged to one project. The new rule allows many projects per employee and many employees per project. Show the revision.

Your answer

Unit 4 Developing and reviewing ER models

Revision selection Original supplementary material by Lefa Malatji

4.1 Unary relationships

Model an employee supervised by at most one other employee, and prevent self-supervision conceptually.

Your answer

4.2 Ternary relationships

A supplier supplies a product to a project with an agreed quantity. Explain why three separate pairwise relationships may lose meaning.

Your answer

4.3 Review business rules

A library book title can have many physical copies. Each copy can have many historical loans but at most one current loan. Propose entities and an enforcement note.

Your answer

Unit 5 Normalisation to 3NF

Revision selection Original supplementary material by Lefa Malatji

5.1 Partial dependencies

Normalise Line(OrderID,ProductID,OrderDate,ProductName,Quantity) assuming one line per product per order.

Your answer

5.2 Transitive dependencies

Normalise Employee(EmployeeID,EmployeeName,DepartmentID,DepartmentName) when each department has one name.

Your answer

5.3 State assumptions

Does a table with a single-column primary key automatically satisfy 3NF? Explain with an example.

Your answer

Unit 6 SQL commands and queries

Revision selection Original supplementary material by Lefa Malatji

6.1 NULL aggregates

For marks 40, NULL and 80 calculate COUNT(*), COUNT(Mark), AVG(Mark) and SUM(Mark). Explain the integer-average issue in SQL Server.

Your answer

6.2 Filtering groups

Write SQL listing CourseID and average Mark for courses with at least five marks of 50 or above. Include only those passing marks in both the average and the count.

Your answer

6.3 Safe changes

You need to delete one test student. Give a safe sequence and explain the role of foreign keys.

Your answer